

A Mathematical and Simulation Study of Lanthanum-Modified PZT for Enhanced Serendipity Shaped Cantilever Piezoelectric Energy Harvester

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Piezoelectric energy harvesting is widely explored for powering low-frequency structural-health-monitoring (SHM) systems. The effect of lanthanum (La) mixing on the performance of PZT-5A and PZT-5J, particularly in nonuniform cantilever geometries, remains insufficiently explored. In this work, the effect of La concentrations of 1.12, 2.24, and 3.36 percentage of weights in a serendipity shaped cantilever piezoelectric energy harvester were examined using copper, silicon, and structural-steel substrates with nickel electrodes. A reduced single-degree-of-freedom electromechanical model was developed to interpret the simulated behaviour. Results show that La mixing enhances output voltage and harvested power across all configurations. The highest performance was obtained for PZT-5A with structural steel and 3.36 % La, yielding 15.47 V and 9.98 mW at 32 Hz. For PZT-5J with 3.36 % La on copper, 13.466 V and 7.55 mW were achieved. Analytical calculations predicted a maximum power of 9.319 mW at an optimal load of 50 k Ω . Capacitance of 96.2 nF for PZT-5A and 107.5 nF for PZT-5J corresponded to electrical reactance near 51.7 k Ω and 46.3 k Ω . An extraction efficiency of 26.1 % under 2 g excitation was estimated. Overall, La mixing improved the output by 12–13 %, providing an effective design pathway for enhanced PZT-based energy harvesters for low-frequency SHM applications.

Keywords: Energy harvester, Piezoelectric, Lanthanum, Nickel, PZT-5A, PZT-5J

1 Introduction

Over the last decade, there has been a significant rise in the development and implementation of energy-harvesting technologies that focus on providing continuous power for low-power electronic devices. These methods are becoming more prevalent in wireless data transmission, sensing, actuation, and healthcare monitoring systems. In this context, there has been a lot of research into improving the performance of piezoelectric materials, which can transform mechanical vibrations into electrical energy and vice versa. According to research, the majority of ambient vibration sources of practical interest are at low frequencies, often less than 200 Hz. Piezoelectric energy harvesters (PEHs) are ideal for such situations due to their high power density, simple structural design, and ease of integration^{1,2}. Among available piezoelectric ceramics, PZT-5A and PZT-5J are widely used because of their high piezoelectric coefficients, thermal stability, mechanical robustness, and cost-effectiveness. The present work focuses on improving their electromechanical performance through lanthanum mixing.

Lanthanum mixing has been extensively explored in various piezoelectric materials to enhance their energy harvesting performance. In ZnO-based Nano generators, La doping in the range of 0–7.5% has been optimized, with contents above 2.5 % yielding a three- to four-fold increase in output³. Lanthanum-doped barium stannate, a refractory near-zero-index material, has demonstrated strong potential for near-infrared thermal emission and has shown thermal stability at temperatures exceeding 1000 °C⁴. Similarly, Nano-sized lanthanum-modified bismuth titanate (BLT) powders prepared via high-speed ball milling have exhibited more than four times the harvested energy compared to their micro-sized counterparts⁵. ZnO nanorods doped with lanthanum have also been reported to produce nearly three times the output of intrinsic piezoelectric nanogenerators⁶. In addition, piezoelectric energy-harvesting systems have been integrated into building envelopes to power fiber-grating sensors, enabling continuous structural-health monitoring⁷. Studies on perforated rectangular cantilevers have further shown increased power extraction from wind and structural vibrations, supporting applications in wireless monitoring frameworks⁸. Flexible composite systems such as

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